TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REGISTER

Class: **Duration time**: from 09/2026 To 03/2027

(*) Profession: Software Engineer **Specialty**: SE

(*) Kinds of person make registers: ☐ Lecturer ☒ Students

1. Register information for supervisor (if have)

No.	Full name	Phone	E-Mail	Title
Supervisor	Thân Thị Ngọc Vân		vanttn@fpt.edu.vn	Giảng viên

2. Register information for students (if have)

No.	Full name	Student code	Phone	E-mail	Role in Group
1	Lê Thành Minh Trung	SE183752		trungltmse183752@fpt.edu.vn	Leader
2	Nguyễn Xuân Lộc	SE183782		locnxse183782@fpt.edu.vn	Member
3	Trần Võ Quang Duy	SE182146		duytvqse182146@fpt.edu.vn	Member
4	Võ Minh Luân	SE182794		luanvmse182794@fpt.edu.vn	Member
5					

3. Register content of Capstone Project

(*) 3.1. Capstone Project name:

3.1.1. English: ***CampusStay FPT – Dormitory Registration and Room-Scheduling Management System for FPT University's Tân Uyên Campus, Bình Dương***

3.1.2. Vietnamese: ***CampusStay FPT – Hệ thống quản lý đăng ký ký túc xá và sắp xếp phòng tại cơ sở Tân Uyên (Bình Dương) của Đại học FPT.***

(*) 3.2. Main proposal content (including result and product)

a) Context:

The FPT Education Complex in Thái Hòa Ward, Tân Uyên (Bình Dương, part of Ho Chi Minh City since the 2025 provincial merger) broke ground on 18 June 2025 on a site of over 84,000 m² with roughly 130,000 m² of floor area, including lecture halls, dormitories, multi-functional areas and technology learning spaces, and is planned to serve up to 4,000 learners. As a brand-new campus it has no dormitory system yet: no legacy process to migrate, but nothing to run registration, room and bed assignment, payments, check-in and check-out, Term transitions, maintenance and family visits once the dormitories open.

Because the dormitory is not operating yet, the requirements are based on (1) how FPT University's operating dormitories work (registration per Term, Deposit and Balance, room types and classes, Block staff, check-in and check-out), (2) the documented process of VNU-HCM Dormitory Zone B (Khu B), compared with FPT's model on Terms, programmes and fees, (3) the Ministry of Education's dormitory regulation (Circular 27/2011/TT-BGDĐT), and (4) the proven scope of student-housing platforms such as StarRez (self-service check-in and check-out, visitor management). Every business value (prices, deposit percentage, windows, eligibility, matching criteria, refund and termination rules, time limits, event capacity) is configuration, so the system can follow the dormitory's real rules when it opens.

CampusStay FPT is not a replacement for FPT University's student information system: the student list and the enrolled list of each Term are imported from files and reconciled. It owns the dormitory workflow: registration periods, Applications, assignment, Invoices and payment confirmation, residence (door codes, self-service check-in and check-out with a room-condition report), Term transitions of continuing residents, requests and maintenance, family events and visitors, notifications, an AI assistant and reports. It does not process tuition fees.

b) Proposed Solutions:

A web app for students (phone-first) and for the dormitory's Staff, Managers and Administrator, an Android app for students, and one REST API with a PostgreSQL database. Students choose a Room Type and a Room Class; a rule-based batch engine places them fairly. Payments are VietQR transfers to the dormitory's account that students claim and Managers confirm. Students check in and out by themselves with a digital room-condition report that the system compares with the baseline recorded by Staff. Managers organise Family Days on which students' relatives, registered in advance, visit the common areas with an individual visitor pass checked by Staff. An AI chatbot runs on a local language model and answers only from the system's own data.

System actors:

1. Student (applicant and resident)
    - Apply for a Term, invite a roommate (2-bed rooms), pay by VietQR and claim the transfer
    - Check in and out by themselves with the room-condition report; see the Room, door code, check-in pass and Invoices
    - Submit requests of every type (maintenance, late check-in, room change, swap, Room Class change, extension or end of stay, check-out); register relatives for a Family Day; ask the AI chatbot
2. Staff (per Block)
    - Record the room baseline before check-in; verify self-service check-in and check-out reports; meter readings, door codes, furniture and inspections
    - Handle the requests of the Block; verify Family Day visitors and check them in and out at the reception
3. Manager (dormitory manager, every Block; added after Review 1)
    - Request registration periods; verify student accounts; import and reconcile student lists
    - Review Applications, run and approve the assignment, confirm payments, decide late check-ins, terminations and refunds
    - Create, publish, cancel and close Family Days and decide their registrations; follow dashboards, reports and overdue requests; set routing rules, time limits and Block settings; manage Staff accounts
4. Administrator
    - Build and publish Terms from the Managers' requests; configure prices, rules, users, notification texts and the chatbot's knowledge base

External actor: Family Visitor (a student's relative, no account)

- Is registered by the student, shows the visitor pass and an identity document at the reception, stays in the allowed common areas and checks out when leaving

The four roles stay unchanged; new students and parents may also ask the chatbot public questions without signing in.

c) Functional Requirements:

Student. The Student shall be able to:

- Sign in with a password or the FPT Google account; a new Google account is verified by a Manager before applying

- Apply for a Term, with eligibility checked, choosing Room Type, Room Class, roommate (2-bed) and stay length (number of Terms), and accept the house rules
- Pay the Deposit, Balance, next-Term fee and Utility Invoices by VietQR transfer and claim the transfer
- See the assigned Room and Bed, roommates, furniture, the door code (once paid and from the check-in start) and the check-in pass
- Complete self-service check-in with a digital room-condition report (personal items of the Bed, shared items of the Room, meter readings, photos) and a commitment that it is true; start self-service check-out with the same report
- Submit and follow requests in one module by type: maintenance, late check-in, next-Term room change, swap, Room Class change, extension or end of the stay, check-out; answer a Round-2 offer
- Register relatives for a Family Day (time slot, name, relationship, contact), up to the configured number; follow the registration, see one visitor pass (QR) per relative and cancel
- Ask the AI chatbot about the dormitory and their own Room, Invoices and requests
- Receive notifications in the app, by push (Android) and by email

Dormitory Staff. The Staff shall be able to:

- Record the baseline of a Room before check-in: condition of shared and personal items, meter readings, photos
- Verify self-service check-in and check-out reports: re-inspect on a mismatch, correct the baseline or hold a joint session with the student; settle damage per furniture item and the Utility Deposit
- Set and confirm door codes, manage furniture (shared and personal items) and the Block store, run inspections
- Enter or import meter readings and close the month (Utility Invoices split by days lived)
- Handle the requests of their Blocks within their time limits; move residents for operational reasons
- Scan visitor passes, verify the visitor's identity, check visitors in and out, follow the visitors inside and handle exceptions (late arrival, invalid pass, overstay)
- Post announcements to the residents of their Blocks

Manager. The Manager shall be able to:

- Request registration periods; verify accounts; import the student list; reconcile the enrolled list
- Review Applications; run the batch assignment, resolve conflicts and approve it; run Round 2
- Confirm or reject payment claims; issue next-Term fees; decide refunds, late check-ins and terminations
- Create and publish Family Days (date, registration period, time slots, capacity, allowed areas, visitors per student); approve or reject registrations; follow the live visitor dashboard; close the event and read its report
- Set routing rules, time limits, Block settings, Room Classes by Term and inspection schedules; manage Staff accounts
- Follow dashboards, the Term dashboard, reports and overdue requests; export every list

Administrator. The Administrator shall be able to:

- Build and publish Terms (timeline, Rooms, prices, rules, information) from the Managers' requests
- Configure Room Types, Room Classes, fees, Utility, eligibility, priority groups, matching criteria, deadlines, refunds and stay rules
- Manage Blocks, Rooms, furniture types, users and roles, notification texts, email campaigns and the chatbot's knowledge base; read the Audit Log

Assignment engine and automation. The system shall automatically:

- Place approved students by gender, Room Type and Class, roommate pairs, priority groups and matching criteria; flag conflicts; give the same result for the same input
- Compare a check-in report with the baseline and record the check-in when they match
- Expire unpaid Deposits, forfeit unpaid Balances, release door codes, run the requests-window milestones, warn and terminate for Utility debt, escalate overdue requests, check Family Day capacity, issue and revoke visitor passes, mark no-shows, flag visitors not checked out and send reminders
- Answer chatbot questions with a local language model that reads the system's data only through read-only functions limited to what the user may see

d) Non-Functional Requirements:

- Security and privacy: access by role and Block; passwords hashed, door codes encrypted; the chatbot sees only the asking user's data and sends nothing to external AI services; visitor data kept only as needed, never an ID number in a visitor pass; personal data processed according to the Personal Data Protection Law 91/2025/QH15.
- Performance and capacity: p95 response ≤ 1 s with 200 concurrent users on 3,680 Beds and 6,200 students; the chatbot starts answering within 5 s on the production server.
- Integrity: a Bed is never given twice in a Term; money of an Invoice never changes once issued; a check-in is recorded only once per Allocation or visitor; a Family Day slot never exceeds its capacity.
- Availability: the API answers its health check in at least 99 % of 5-minute checks during the demo period.
- Configurability: Terms, prices, rules, time limits and event settings change without a new release.
- Usability: Vietnamese and English; student portal usable on phones.
- Deployability: every change that passes its tests is deployed automatically; rollback in minutes.

e) Theory & Practical:

Documents: Reports 1–7. Technologies:

- React (web app for all roles), Flutter (mobile app for students, built for Android)
- Spring Boot (Java 21) REST API, PostgreSQL with Flyway migrations
- A rule-based batch room-assignment engine
- A local language model (Qwen, served by Ollama) with tool calling for the chatbot
- Firebase Cloud Messaging, SMTP email, in-app notifications
- VietQR transfer images; QR passes; Google sign-in for FPT accounts
- Docker, GitHub Actions CI/CD, Vercel and Azure hosting

f) Products (Expected Deliverables):

A web app for students and for the dormitory's Staff, Managers and Administrator; an Android app for students; an AI chatbot for students (and a public mode for new students and parents); Family Day management; the REST API with its database scripts and automated tests; Reports 1–7.

g) Proposed Tasks:

- P1 Literature review: student-housing platforms, Vietnamese dormitory practice and regulation (VNU-HCM Zone B), FPT's dormitory model.
- P2 Requirement analysis: registration, assignment, payments, residence, Term transition, requests, maintenance, Utility and family-event workflows.
- P3 System design: architecture, database, REST API, state machines, the assignment engine, the chatbot's tools.

- P4 Registration and payments: registration periods, Applications, assignment, VietQR claims and confirmation.
- P5 Residence and operations: door codes, self-service check-in and check-out, Term transition, requests and maintenance, Utility and debt, Family Day.
- P6 Clients, AI and reports: web app for four roles, Android app, the AI chatbot, dashboards and exports.
- P7 Testing and documentation: automated and system tests, user acceptance, reports and demonstration.

4. Other comments (propose all relative things if have):

Version 2.1, proposed after Review 2 (31/10/2026) and the supervisor meeting (05/11/2026); it replaces version 2.0 proposed at Review 2. The changes from version 1.0 and their reasons are listed in the attached Record of Changes (BM03); changes 5 and 13–17 are new or revised in 2.1.

HCM, date 08/11/2026

Supervisor (If have)	**On behalf of Registers**
(Sign and full name)	*(Sign and full name)*

Registration Form – Record of Changes

1. Project Information

Project code	FA26SE260
Project name	CampusStay FPT – Dormitory Registration and Room-Scheduling Management System for FPT University's Tân Uyên Campus, Binh Duong
Team	CampusStay FPT — Lê Thành Minh Trung (Leader), Nguyễn Xuân Lộc, Trần Võ Quang Duy, Võ Minh Luân
Supervisor	Thân Thị Ngọc Vân
Current registration version	1.0 (28/08/2026)
Proposed version	2.1 (08/11/2026): after Review 2 on 31/10/2026 and the supervisor meeting on 05/11/2026; replaces the 2.0 proposed at Review 2, on which the reviewers recorded no decisions
Version	2.1
Date	08/11/2026

2. Changes

☐No changes to the current registration form

No.	Section	Current content	Proposed content	Reason	Proposed by
1	2. Students	No members listed	4 members with student codes, emails and roles (phones omitted)	Team formed for the capstone	Lê Thành Minh Trung
2	3.2 a) Context	Tân Uyên City, Bình Dương; scope adapted from StarRez	Tân Uyên, part of Ho Chi Minh City since 2025; basis of the requirements: FPT's operating dormitories, Circular 27/2011/TT-BGDĐT, StarRez; every business value configurable	Review 1 feedback (Q1–Q3: how to specify a dormitory that is not operating yet); 2025 administrative merger	Lê Thành Minh Trung
3	3.2 b) Actors	Student, Dormitory Staff / Residence Life Officer, Administrator	Student; Staff per Block; Manager for every Block (reports, statistics, decisions, Block management); Administrator	Review 1 feedback (Q4: add the Manager role)	Lê Thành Minh Trung
4	3.2 c) Student: room selection	Browse and select a room/bed; search and request a roommate	Choose Room Type and Room Class; invite a roommate for 2-bed rooms; the batch	Iteration 1: fair placement with gender and Block rules and no race for	Lê Thành Minh Trung

No.	Section	Current content	Proposed content	Reason	Proposed by
			engine assigns the Bed	a Bed; the complexity moves into the engine	
5	3.2 c) Student: contract, check-in/out	Digital housing contract; self-service check-in/check-out with a digital room-condition report	Consent to the house rules recorded instead of an e-contract. Self-service check-in and check-out kept: Staff records a baseline of the Room (shared and personal items, meter readings, photos) before check-in; the student fills the room-condition report and confirms a commitment that it is true; the system records the check-in when the report matches the baseline, otherwise Staff re-inspects and a joint session follows. Check-out is started by the student and verified by Staff. (Replaces the desk check-in proposed in v2.0.)	Review 2 (31/10/2026) and supervisor meeting (05/11/2026): the self-service item of v1.0 must be kept; only the e-contract is dropped (no legal e-signature provider)	Lê Thành Minh Trung
6	3.2 c) Payment	Pay via a VietQR-compatible payment gateway	VietQR transfer to the dormitory account; the student claims it, a Manager confirms or rejects it	Constraint from partners: no bank API or gateway contract for a student project	Lê Thành Minh Trung
7	3.2 c) Communication	In-app notification, email or SMS	In-app notifications, push to the Android app, email; announcements board	SMS gateway cost; push and email cover the cases	Võ Minh Luân
8	3.2 c) Added functions	—	Registration-period requests and the Term wizard; Term transition (stay length, requests window M2–M7, next-Term fees); late check-in; Round 2; monthly Utility billing and debt; door codes; furniture	Iteration 1: a dormitory runs across Terms; these add business complexity	Nguyễn Xuân Lộc
9	3.2 c) SIS integration, forecasting	Integrate with the SIS; forecast occupancy and vacancy	Import of the student list and reconciliation of the enrolled list per Term; occupancy and Term dashboards	Constraint from data: the SIS is not accessible; a new campus has no history to forecast from	Lê Thành Minh Trung
10	3.2 d) Non-functional requirements	Decree 13/2023/NĐ-CP; capacity 4,000 learners; SIS and gateway interoperability	Law 91/2025/QH15; measurable NFRs: p95 ≤ 1 s at 200 users, integrity, availability, two languages,	The Law replaces the Decree from 01/01/2026; NFRs need metric,	Võ Minh Luân

No.	Section	Current content	Proposed content	Reason	Proposed by
			deployability; chatbot privacy and response time	threshold and method	
11	3.2 e), f) Technologies, products	ReactJS or Flutter/React Native; Spring Boot or Node.js; mobile/web app, staff web app, admin dashboard	React web app for all roles; Flutter mobile app for students (built for Android); Spring Boot API; local language model for the chatbot; Docker, GitHub Actions, Vercel, Azure	Team skills; Flutter covers push and Google sign-in on Android; one web app for all back-office roles. Corrects v2.0, which named Kotlin / Jetpack Compose	Trần Võ Quang Duy
12	3.2 g) Proposed tasks	P1–P7 around room selection and a payment gateway	P1–P7 updated to the scope above	Follows the other changes	Lê Thành Minh Trung
13	3.2 b), c) Family Day (module VIS)	— (v1.0 context: front-desk and visitor management of StarRez)	Family Day in the common areas: a Manager creates and publishes the event (registration period, time slots, capacity, allowed areas, visitors per student) and decides registrations; a resident registers relatives; each approved relative gets an individual visitor pass (QR); Staff verify the identity document and check visitors in and out at the reception; the system marks no-shows, flags visitors not checked out and reports the event. Family Visitor is an external actor without an account; no visits to student rooms, no overnight stay, no ID number in a pass.	Review 2 (31/10/2026) and supervisor meeting (05/11/2026): supervisor asked for family visits with mail notification, registration and gate control	Võ Minh Luân
14	3.2 b), c), d), e) AI chatbot	—	AI chatbot for students (signed in) and, in a public mode without an account, for new students and parents. It runs on a local language model (no external AI API) and reads data only through read-only functions limited to what the asking user may see; no new role.	Review 2 (31/10/2026) and supervisor meeting (05/11/2026): keep the chatbot for new students and parents; local model and data permissions demonstrated	Võ Minh Luân
15	3.2 c) Requests	Submit a room-change or maintenance request and track its status; process room-change requests and transfers	One request module by type (maintenance, late check-in, room change, swap, Room Class change, extension or	Review 2 (31/10/2026) and supervisor meeting (05/11/2026): same	Nguyễn Xuân Lộc

No.	Section	Current content	Proposed content	Reason	Proposed by
			end of stay, check-out): one lifecycle, one list for the student, one queue for Staff and Managers; each type keeps its own rules	functions, organised as one module by request type	
16	3.2 c) Staff: furniture	Manage building/floor/room/bed inventory	Furniture split into shared items of the Room and personal items of each Bed; each student confirms his or her personal items at check-in, the basis for damage charges at check-out	Review 2 (31/10/2026) and supervisor meeting (05/11/2026): clear responsibility per student for compensation	Trần Võ Quang Duy
17	3.2 a) Context: basis	— (v2.0: practice of VNU-HCM dormitories in general)	Basis of the requirements adds the documented process of VNU-HCM Dormitory Zone B (Khu B), compared with FPT's model on Terms, programmes and fees	Review 2 (31/10/2026) and supervisor meeting (05/11/2026): supervisor asked for a documented reference process and comparison	Lê Thành Minh Trung

3. Decision at Review 2 (completed by reviewers)

No.	Decision (Accepted / Adjusted / Rejected)	Reviewer note
1		
2		
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No.	Decision (Accepted / Adjusted / Rejected)	Reviewer note
14		
15		
16		
17		